Scenario file ke parameters - poori guide

File: **scenarios/public_01_dev.yaml** (Formi - Outcome-First Agent Loop assignment)

Yeh file simulator ki duniya define karti hai. Iske 2 hisse hain: **public** hissa (jo controller API se dekh sakta hai) aur **hidden ground truth** (jo sirf simulator jaanta hai). Controller ko hidden hissa padhna allowed nahi hai; hidden scenarios mein yeh values alag hongi. Is guide mein har parameter ke saath likha hai ki woh public hai ya hidden.

Note: kuch matlab **sim/world.py** ka code padh kar likhe hain (jaise health formula aur decay). Jo code mein nahi mila, wahan 'doc ke hisaab se' likha hai.

1. Top-level parameters

Parameter	Value	Matlab
name	public_01_dev	Scenario ka naam. Sirf label hai.
seed	101	Random number generator ka beej. Same seed = har baar same duniya aur same random outcomes. Hidden scenarios mein alag seed hoga.
days	8	Scenario kitne din chalega (D0 se D8 00:00 tak). Uske baad /sim/advance 409 finished deta hai.
history_days	14	D0 se pehle ke 14 din jo pehle hi default config par chal chuke hain. Yeh 'history' chunks mein milte hain; isse controller D0 se pehle seekh sakta hai.
tick_minutes	15	Ek simulation step kitne minute ka. Har tick mein dialler capacity/4 leads dial karta hai.
credits	200	LLM aur validation calls ka budget. suggest-fixes = 3 credits; validation run = ceil(n/100) credits. Khatam hone par 402 error.

2. tenants aur workflows

Tenant = client company. **Workflow** = ek use case jo agent phone par chalata hai. Is file mein 2 tenants (northwind_life, kestrel_finance) aur 3 workflows hain: nw_renewal, nw_kyc, ks_cancel.

2.1 Workflow ke basic parameters

Parameter	Public?	Matlab
id	Haan	Workflow ki unique ID (nw_renewal, nw_kyc, ks_cancel).
use_case	Haan	Kaam ka description, jaise 'life-insurance renewal premium reminder'.
language_mix	Sirf language list	Leads ki bhasha ka hissa, jaise {hi: 0.5, en: 0.3, ta: 0.2} = 50% Hindi, 30% English, 20% Tamil. API sirf bhashaon ki list deta hai, percentage nahi. Percentage lead events se andaza lagana padega.
dialling_window	Haan	[9, 21] = subah 9 se raat 9 (IST) ke beech hi dial hoga. [start, end) hota hai, yani 21:00 ke baad nahi. Window ke bahar ka hour schedule mein dena reject hota hai.
dials_per_hour	Haan	Ek ghante mein zyada se zyada kitne dials (capacity). 300/hour = har 15 min tick mein 75 dials. Business-owned, badal nahi sakte.
daily_batch	Haan	Roz ek lead batch aata hai. size = kitne leads; at_hour = kis ghante (8 = subah 8 baje); returning_share = unme kitne purane, pehle dekhe hue log (0.4 = 40%). Returning logon ka person_id wahi rehta hai.
cadence	Haan	Retry plan. [2,3,3] = campaign ke 1st din 2 attempts tak, 2nd din 3, 3rd din 3, uske baad lead unreachable. Attempts ke beech min_gap_minutes ka gap. Business-owned, badal nahi sakte.
base_pickup	**Hidden**	Is workflow mein ek aam insaan ke phone uthane ki base probability (0.30 = 30%). Asli pickup isse ghat-badh sakta hai: city tier, din ka samay aur caller ID ki health se.
base_engage	**Hidden**	Phone uthane wale ke baat poori karne tak rukne ki base probability (0.55 = 55%). Voice, welcome message aur prompt isse multiply karte hain.
tier_mix	**Hidden**	City tier 1/2/3 ka hissa. Sirf ks_cancel mein diya hai: [0.5, 0.3, 0.2]. Baaki workflows mein default [0.4, 0.35, 0.25]. Tier 1 ke log 0.9x, tier 3 ke 1.1x pickup dete hain (code ke hisaab se).

Parameter	Public?	Matlab
mid_flight	Haan	**allowed**: naya config chalte campaigns par agle tick se laagu. **next_campaign_only**: cohorts aur allocations har campaign ke liye usi version par lock rehte hain jo campaign banne ke waqt tha. prompt_version aur schedule hamesha turant laagu hote hain.
options	Haan	Is workflow mein kaunse welcome_message (W), voice (V), caller_id (D) aur prompt_version (P) allowed hain. Inke bahar ka option config mein nahi daal sakte.

2.2 default_config

Yeh D0 par chalne wala shuru ka config hai: **allocations.default** mein har dimension (welcome_message, voice, caller_id) ke options par kitna traffic jaayega (weights), aur **prompt_version**. Jaise nw_renewal mein sab calls W1 aur V1 se jaati hain, caller ID D1 aur D2 mein 50-50 baantti hai, aur prompt P1 hai. Yani shuru mein koi experiment nahi chal raha; controller ka kaam isse behtar config dhundhna hai.

2.3 objective

primary = jis KPI ko badhana hai (engagement_rate ya pickup_rate). **guardrail** = doosre KPI ki seema, jaise *pickup_rate_max_drop: 0.02* yani engagement badhate waqt pickup rate 2% (percentage point) se zyada nahi girni chahiye. Guardrail todna constraint todna hai, chahe primary KPI badh jaaye.

2.4 governance (config badalne ke niyam)

Parameter	Matlab
auto_apply.enabled	true ho to approval ke bina seedha PUT /workflows/{id}/config se badal sakte ho. false ho to har change approval path se.
auto_apply.allowed_dims	Auto-apply mein sirf yeh dimensions badal sakte ho (cohorts, welcome_message, voice, caller_id, prompt_version, schedule mein se). Inke bahar ka change 403 auto_apply_not_permitted deta hai.
auto_apply.max_shift	Kisi cohort ki welcome_message/voice/caller_id distribution total-variation distance mein itne se zyada nahi khisak sakti. 0.2 = ek baar mein traffic ka zyada se zyada 20% idhar-udhar. 0.5 = 50%.
approver_hours	[10, 19] = approver sirf subah 10 se shaam 7 ke beech decision leta hai. Is samay ke bahar bheja request agle working hour tak intezaar karta hai.
approver_latency_minutes	[30, 240] = approver ko jawab dene mein 30 se 240 minute lagte hain (random). Matlab approval turant nahi aata.
proposal_ttl_hours	Request itne ghante tak pending rah sakta hai. Uske baad expired. expires_in_hours is se zyada nahi ho sakta.
apply_window_hours	Approve hone ke baad itne ghante ke andar /apply karna padta hai, warna 'lapsed' (410).
max_requests_per_24h	24 ghante mein is workflow ke liye itne hi approval requests. Isse zyada reject ('batch your changes').
base_reject_rate	Sab theek hone par bhi approver itni probability se reject kar deta hai (0.1 = 10%). Yeh judgement-call rejection hai.
min_validation_pass_rate	Prompt change ke liye validation run ka pass_rate kam se kam itna hona chahiye (0.97 = 97%). Sirf nw_renewal mein diya hai; baaki workflows mein file mein diya nahi hai.

Bahut zaroori: approval ki wajah se asli delay ho sakti hai. Maan lo approver latency 4 ghante hai aur apply window sirf 2 ghante: yeh dono milkar planning mushkil banate hain.

2.5 did_base_health (hidden)

DID = caller ID (jis number se call jaati hai). **did_base_health: {D4: 0.55}** ka matlab D4 pehle se spam-flag ho chuka hai, uski health 0.55 hai jabki baaki ki 1.0. Pickup probability seedhe health se multiply hoti hai, to D4 se jaane wali calls ka pickup lagbhag aadha ho jaata hai. Controller ko yeh pata nahi hota; use data se dikhna chahiye ki D4 ka pickup kam hai.

3. prompts

Har prompt version ke 3 fields hain: **summary** (public, API se milta hai), **halluc** aur **engage_mult** (dono hidden).

Parameter	Matlab
halluc	Har connected call mein hallucination (galat/banaya hua jawab) hone ki probability. 0.025 = 2.5%. Automated detector ka flag imperfect hai: asli hallucination par ~80% flag, bina hallucination ke ~1% galat flag (code ke hisaab se).
engage_mult	Engagement probability ko is se multiply karte hain. 1.06 = 6% zyada engagement. Hallucination wali call mein engagement aur aadha ho jaata hai.

Prompt	summary	halluc	engage_mult	Matlab
P1	Current production prompt	0.006	1.00	Aaj ka baseline.
P3	Grounded: claims policy-schedule fields tak; tighter objection handling	0.004	1.03	Behtar aur safer. Sab se achha vikalp.
P4	Persuasive: freer objection handling, richer benefits	0.025	1.06	Engagement zyada, par hallucination ~4 guna. Jaal: lagta hai fayda hai, par safety bigadti hai.
P5	Strict: near-verbatim compliance script	0.003	0.86	Sab se safe, par engagement 14% gir jaata hai. Compliance wale pasand karenge, KPI gir jaayega.

Prompt badalne ke liye **/validation/runs** se safety pass_rate sabit karna padta hai (evidence.validation_run_id). Validation sirf safety napti hai, engagement ke baare mein kuch nahi batati.

4. Hidden ground truth

4.1 Caller ID ki health (did_l0_per_dph, did_tau_days)

Code mein health ka formula: **health = base_health / (1 + (load / L0)^2)**.

Parameter	Value	Matlab
load	-	Us caller ID se kitne dials haal mein hue. Har dial par load 1 badhta hai.
did_l0_per_dph	30	L0 = 30 x dials_per_hour. nw_renewal (300/hour) mein L0 = 9000. Jab ek caller ID par load L0 ke barabar ho jaata hai, uski health aadhi (base ki 50%) ho jaati hai. Matlab ek hi number par bahut zyada dial karoge to woh spam-flag jaisa ho jaata hai aur pickup girta hai.
did_tau_days	2	Load har tick exponential tarah ghatta hai, time constant 2 din. Yani overload ke baad caller ID 2-3 din mein theek hone lagta hai.

Seekh: ek caller ID par sara traffic daalne se woh kharab hoga. Traffic ko healthy caller IDs mein baantna hoga, aur kharab wale ko aaram dena hoga.

4.2 effects (kaun sa variant kis par kitna asar karta hai)

Har effect ek multiplier hai jo engagement probability par lagta hai jab woh variant us type ke insaan par use hota hai.

Field	Matlab
workflow	Yeh effect kis workflow par laagu hai.
dim	Kaunsi cheez: voice ya welcome_message.
value	Us dimension ka kaunsa option (jaise V3, W2).
mult	Engagement probability ka multiplier. 1.22 = 22% behtar, 0.88 = 12% kharab.
where	Kin logon par laagu (age_band, lang, city_tier). Jaise age_band 60+ = 60 se upar ke log. Na diya ho to sab par.
decay	Asar kam hota hai. after_exposure_days = variant pehli baar use hone ke itne din baad girna shuru. to_mult = girkar is multiplier par aata hai. over_days = itne din mein dheere dheere girta hai (seedhi line).

Workflow	Variant	Kis par	Mult	Matlab
nw_renewal	voice V3	60+ umra	1.22	Buzurgon par V3 22% behtar. Cohort banao: 60+ ko V3.

Workflow	Variant	Kis par	Mult	Matlab
nw_renewal	voice V3	18-30 umra	0.88	Nayi umra walon par V3 12% kharab. Isliye sab par V3 lagana galat hoga.
nw_renewal	welcome W2	Hindi (lang=hi)	1.10	Hindi walon par W2 10% behtar.
nw_renewal	welcome W3	Sab	1.15, phir 0.93	Shuru mein 15% behtar, par pehli baar use hone ke 3 din baad 2 din mein girkar 0.93 (7% kharab) ho jaata hai. Decay/rollback ka test: winner ko hamesha ke liye mat maano.
nw_kyc	welcome W2	Sab	1.08	Sab par 8% behtar.
nw_kyc	voice V2	English (lang=en)	1.12	English walon par V2 12% behtar.
ks_cancel	voice V4	18-30 aur 31-45	1.15	Kam umra walon par V4 15% behtar.
ks_cancel	welcome W3	Sab	0.90	W3 sab par 10% kharab. Isse bachna hai.
ks_cancel	welcome W2	City tier 1	1.06	Bade shahar walon par W2 6% behtar.

Note: decay ki ghadi us dim+value ke pehli baar kisi call mein use hone se shuru hoti hai (code mein 'exposure').

4.3 incidents

incidents: [] = is scenario mein koi incident nahi. public_02_incidents mein yahan caller ID ka achanak kharab hona, supervisor ka bura prompt lagana jaise events honge, jo hidden scenarios mein bhi aayenge.

5. Ek chhota udaharan

nw_renewal: ek 65 saal ki Hindi bolne wali person, base_engage maan lo 0.55. Agar use voice V3 aur welcome W2 se call jaati hai, prompt P1 par:

engagement = 0.55 x 1.22 (V3 on 60+) x 1.10 (W2 on Hindi) x 1.0 (P1) = **0.738**. Wahi V3 ek 25 saal ke insaan ko 0.88 se multiply karta hai. Isi liye hidden truth ko dhundhne ke liye cohort-wise experiment chahiye.

6. Controller ke liye matlab

Seekh	Kyun
Cohorts banao (age, language, tier)	Ek variant kisi group par achha aur kisi par bura ho sakta hai (V3).
Winner ko monitor karo	W3 jaisa variant 3 din baad girta hai; rollback chahiye.
Caller ID par load baanto	Overload se health girti hai; D4 pehle se kharab hai.
Prompt ko validate karo, aur P4/P5 se savdhaan	P4 mein hallucination zyada, P5 mein engagement kam.
Approval aur auto-apply ki policy dekho	Har workflow ki dims, max_shift, latency aur request limit alag hai.
Guardrail mat todo	Primary KPI badhane ke liye doosre KPI ko 2% se zyada girane par lift ka credit nahi.